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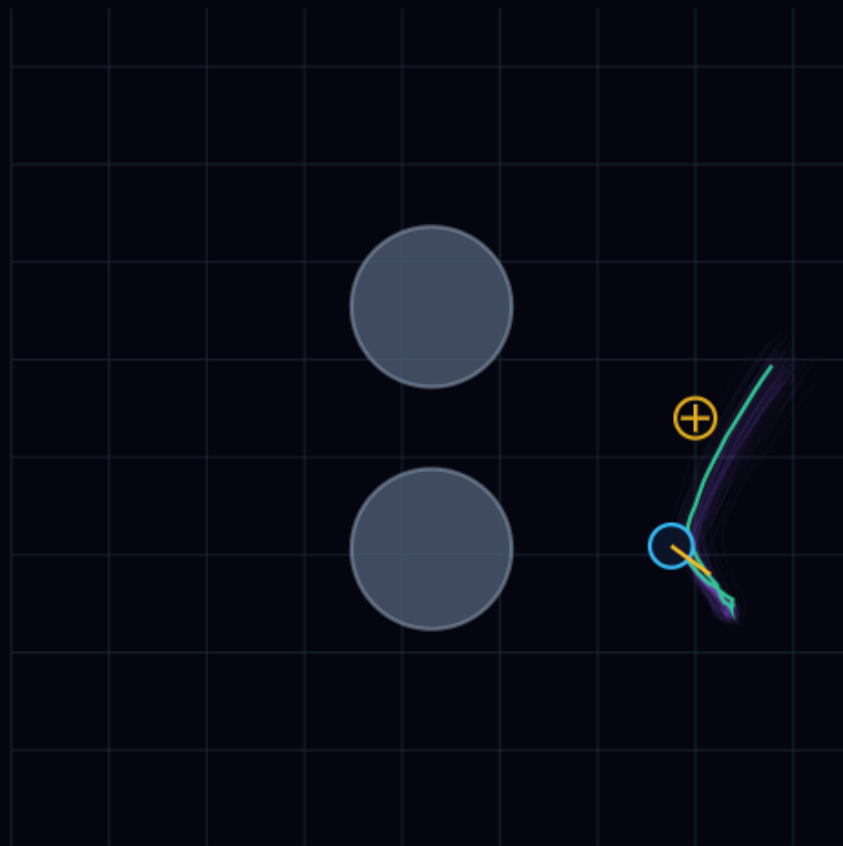
Holonomic 2D Hovercraft MPC (MPPI)

Planar double integrator with l_1 thrust cap: sampling-based MPPI steers to a draggable goal while soft-penalizing circular obstacles — rollout fan and best predicted path drawn live.

Pause

Reset

Click canvas: move goal · MPPI = weighted control averaging

$$u = (0.56, -0.52) \text{ m/s}^2 \cdot |u|_{\max} = 4.0$$


MPPI (HOLONOMIC 2D)

Obstacle layout

Twin disks (corridor)

Horizon H

45

Samples K

84

Noise σ

0.1 m/s²

Temperature λ

0.1

Plan Δt

0.12 s

 l_{null_2} limit

4 m/s²

COST

Position weight

0.5

Velocity weight

0.05

Control effort R

0.01

Terminal weight	40
Obstacle skirt	8
Inside penalty	1200
Skirt length d_0	0.31 m
Velocity damping	1.2
RNG seed	99999

- ✓ Draw rollout fan ($K \leq 140$)

Obstacles are soft costs (exponential skirt + quadratic penetration). Tight gaps need enough q_{crash} and horizon to “see” the corridor.

SHORTCUTS

- Click anywhere in the arena to move the goal (yellow crosshair).
- If the canvas stutters, lower samples K or disable rollout fan.

MEASURED VALUES

Distance to goal	0.656	m
Speed llvl	0.711	m/s
Best rollout cost	46.2	

About this model

This page extends the **sampling-based MPC** idea from **engineering/mpc-pendulum** to a **planar holonomic “hovercraft”** modeled as a **double integrator**: $\ddot{\mathbf{x}} = \mathbf{u}_x$, $\ddot{\mathbf{y}} = \mathbf{u}_y$ with a bounded Euclidean thrust $\|(\mathbf{u}_x, \mathbf{u}_y)\|_2 \leq u_{\text{max}}$.

A **Model Predictive Path Integral (MPPI)** loop draws **K** noisy open-loop acceleration sequences over a horizon **H**, rolls each trajectory forward with simple semi-implicit Euler steps plus light **velocity damping**, and scores rollouts with a **quadratic goal attraction** to a **click-draggable target**, **velocity regularization**, **control effort**, and **soft obstacle costs**

around fixed **circular rocks** (exponential “skirt” when outside plus a strong **quadratic penalty** if the robot disk penetrates the inflated obstacle boundary). The weighted-average control update matches MPPI’s information-theoretic reweighting; the simulator applies the **first mean action** until the next replan at interval Δt . Purple fan trajectories (optional when **K** is not huge) visualize the ensemble; the **green** polyline is the **lowest-cost** sample path from the last plan. Layout presets create a **narrow corridor** or an **offset pillar field** so students can trade **horizon**, **temperature λ** , **noise σ** , and **obstacle weights** against conservatism and compute budget.

Who it's for: Students who finished the pendulum MPPI demo and want the same sampling intuition on a **2D navigation** task before touching full nonholonomic models or QP-based MPC.

KEY TERMS

- Holonomic robot
- Double integrator
- Model Predictive Path Integral (MPPI)
- Trajectory sampling
- Soft collision constraints
- Obstacle avoidance
- Warm-started control sequence

How it works

Holonomic 2D “hovercraft” with $\ddot{\mathbf{x}} = \mathbf{u}_x$, $\ddot{\mathbf{y}} = \mathbf{u}_y$ and bounded **L2** thrust: **MPPI** rolls **K** noisy acceleration trajectories over horizon **H**, adds **obstacle skirts** around circular rocks, and reweights the mean control each step — same sampling spirit as **engineering/mpc-pendulum**, now with **goal seeking** and **collision avoidance** in the plane.

Key equations

$\ddot{\mathbf{x}} = \mathbf{u}_x$, $\ddot{\mathbf{y}} = \mathbf{u}_y$, with $\|(\mathbf{u}_x, \mathbf{u}_y)\|_2 \leq u_{\text{max}}$ – holonomic double integrator (hovercraft abstraction).
MPPI: sample noisy accelerations along the horizon, score rollouts with goal, speed, effort, and obstacle soft penalties, then exponential weights update the mean control sequence.

Frequently asked questions

Why soft obstacles instead of hard “if collision discard rollout”?

Hard rejection creates **discontinuous** costs and brittle gradients for sampling methods—many rollouts die in useless configurations. Smooth penalties keep **grad-like** information in the exponential weights while still exploding cost on deep penetration.

Is this literally a hovercraft dynamics model?

Only as a **cartoon**: real hovercraft have attitude–thrust coupling, fan dynamics, and ground effect. The **double integrator with $\|\mathbf{u}\|_2$ cap** is the standard **holonomic** surrogate used in curricula to isolate **MPC in the plane**.

Why does the robot sometimes scrape obstacles despite MPC?

Soft costs allow **trade-offs**: if **q_pos** dominates or **horizon** is short, the planner may briefly accept small penetration to cut a corner. Raise **q_crash**, lengthen **H**, or increase **R** on control to be more cautious.

Does disabling the rollout fan change the controller?

No—it only hides the **purple** ensemble for performance. The **green** best-path preview and applied **u** are unchanged.

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Sine $\rightarrow$ samples $\rightarrow$ quantization $\rightarrow$ ZOH; Nyquist and optional aliasing demo.